

## Updated Project Charter

**Project Title:** Market Research and Product Validation for Mainland China's DSE Trends and AI Question Generator

**Business Partner:** Master Mock EduTech Limited

### 1. Background of Company and Project

Master Mock EduTech is an educational technology company that has achieved initial success in the Hong Kong market with its AI-based question generation services for the DSE (Hong Kong Diploma of Secondary Education) exam. As the number of mainland students taking the DSE exam grows, the company seeks to assess the feasibility of entering the mainland market, specifically starting in Shanghai and the Yangtze River Delta. This project will migrate the AI analysis framework developed in Semester 1 to analyze mainland social media trends (e.g., XiaoHongShu, Weibo) to provide evidence-led strategic recommendations

### 2. Team: Members, Skills, and Roles

*In this section, introduce your team and how you plan to work together. Fill in the table below.*

### 3. Scope

#### Technical Objective:

This project will deploy an advanced AI-powered tool to systematically collect and analyze real-time social media data (XHS, TikTok, forums) to assess DSE trends and validate demand for AI-assisted learning tools in the Mainland.

**Strategic Objective:** To deliver a decision-ready report focusing on:

1. Trend Insights: Current and projected profiles of mainland DSE students.
2. Demand Analysis: Willingness of schools and tutoring centers to adopt and pay for AI question generation tools.

#### Known Constraints:

**Data Volume & Scalability:** Processing large volumes of content across fragmented digital channels.

**Anti-Scraping & Privacy:** Complying with PIPL while capturing real-time search patterns.

### 4. Key Deliverables

#### Deliverables for This Academic Year (INF101TC):

**Social Media Crawler Script:** A Python-based tool modified to extract DSE-related discussions from platforms like XiaoHongShu.

**EduTech Data Visualization Suite:**

1. Sentiment Score Distribution: Reflecting student anxiety or interest in DSE and AI tools.
2. Keyword Word Clouds: Highlighting the most discussed subjects or exam pain points.

**System Architecture Diagram:** Demonstrating the complete workflow from crawling data from social media to LLM processing, and finally generating AI-driven Market Research Report.

**AI-Generated Strategic Brief (if possible):** A preliminary report generated via LLM API that synthesizes scraped data into market insights.

**Potential Future Deliverables:**

**Advanced AI-Powered Insight System:** A tool integrating social media scraping with automated theme clustering and competitor mapping.

**Final Market Insight Report:** An AI-supported report combining platform-level trends, competitor intelligence, and student need identification to determine "Go/No-Go" for Master Mock.

**Product Refinement Brief & Digital Mock-ups:** Detailed feedback on the current AI Question Generator with visual suggestions (mock-ups) for UI/UX improvements to fit mainland user workflows.

**5. Teacher Comment*****Project Scope***

***This project will deploy an advanced AI-powered scraping, monitoring, and insight-generation tool to systematically collect, structure, and analyse online data related to DSE trends, student behaviour, and exam preparation demand. The tool will scan large volumes of content across Xiao hong shu, TikTok, public websites, forums, education platforms, and other relevant digital channels to capture real-time discussions, search patterns, product signals, and user sentiment.***

***A second focus of the project is to identify and profile Mainland China companies operating in the mock test, IELTS preparation, English learning, and adjacent assessment markets. The AI tool will not only gather company-level information, but also compare offerings, pricing signals, growth patterns, content strategies, customer pain points, and market positioning at scale.***

***Beyond data collection, the AI system will perform automated trend detection, theme clustering, sentiment analysis, competitor mapping, and needs identification. This will allow the project to move beyond descriptive scraping and generate deeper market intelligence on what Chinese students are worried about, what kinds of learning support they actively seek, which formats and services are gaining traction, and where unmet needs may create room for new education products.***

***The final output will be an AI-supported market insight report that combines platform-level trend analysis, competitor intelligence, and student need identification to support opportunity assessment in the DSE and mock test preparation space.***